Concern Resolution Process for In-house Manufacturing

QC Story

QC STORY PROJECT

PROJECT TITLE :-

"To reduce lead time of export model 4WD at Chassis Line"

Mahindra and Mahindra Ltd, Nasik Plant, Scorpio TCF.

	Plan	Actual
Start Date	**1.7.19**	**1.7.19**
Completion Date	**31.12.19**	**25.12.19**

Current Status: **CLOSED**

Cross Functional Team

Sr. No	Description	Name	Designation	Dept.	Phone
1	Team Leader	Rajesh Dhanokar	Manager	IED	9881250515
2	Mentor	Rahul Narkar	HOD	SCTCF	9881080199
3	Member 1	Suyabhan Sor	Dy Manager	Sc TCF	9763003635
4	Member 2	Ajit Choudhry	Dy Manager	Sc TCF	7588616388
5	Member 3	Pravin Patil	Manager	Sc TCF	8888897850
6	Member 4	Hari Nagar	Manager	Sc TCF	9881786767

Step1. Select Subject

Brief description of Problem:-

"To reduce lead time of export model 4WD at Chassis Line"

What : High lead time for export model

Where : At STCF

When : Since last Nov-18.

Who : Top Management

WHY : Offline Assembly

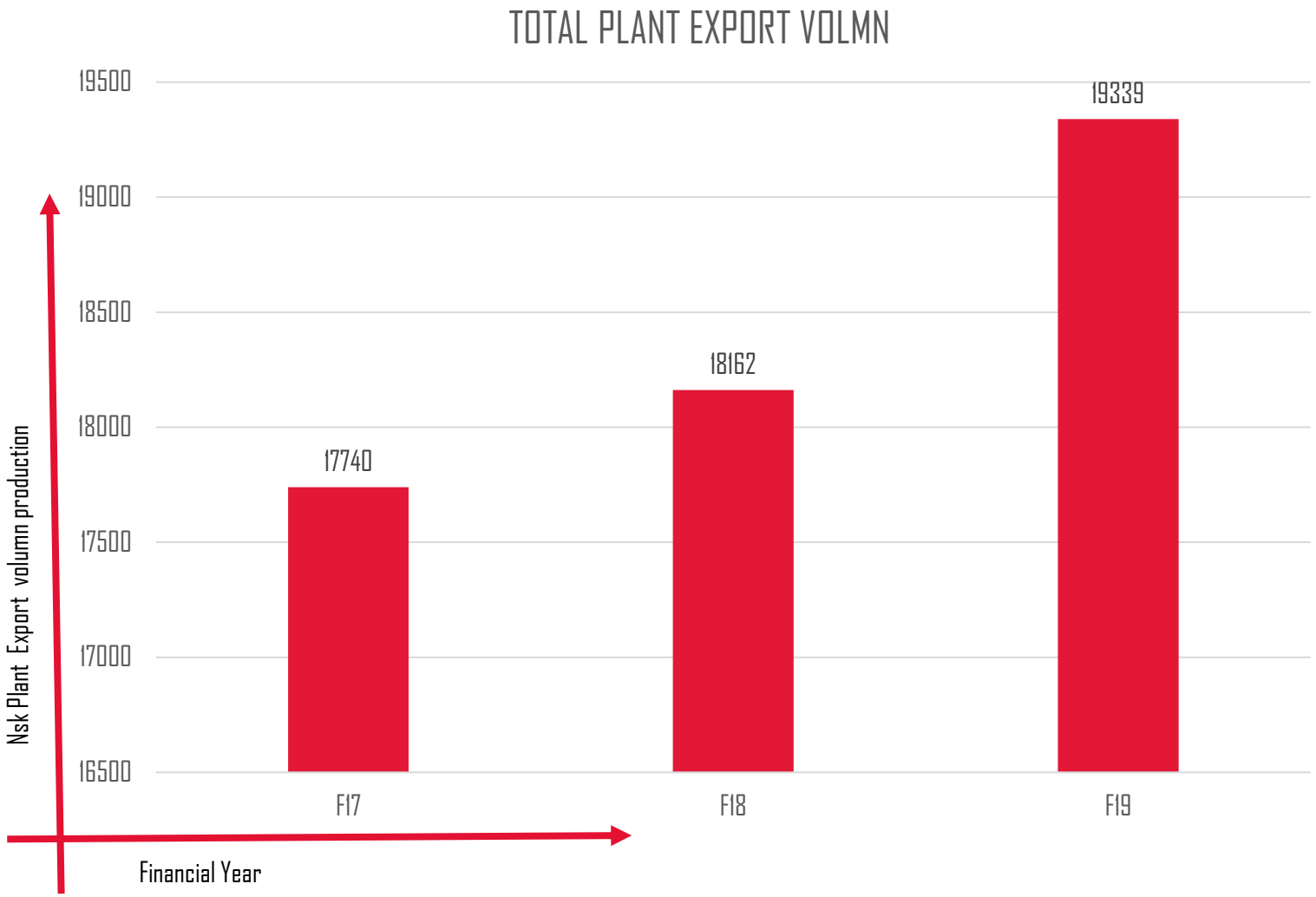
HOW : Excess lead time

2. Justify the choice (reasons)

Overseas Market

Departmental Objective :-
To reduce export lead time

2. Justify the choice



YOY Trend shows increase in volume , which needs to be focus for Productivity improvement

Step 3. Understanding Current Situation

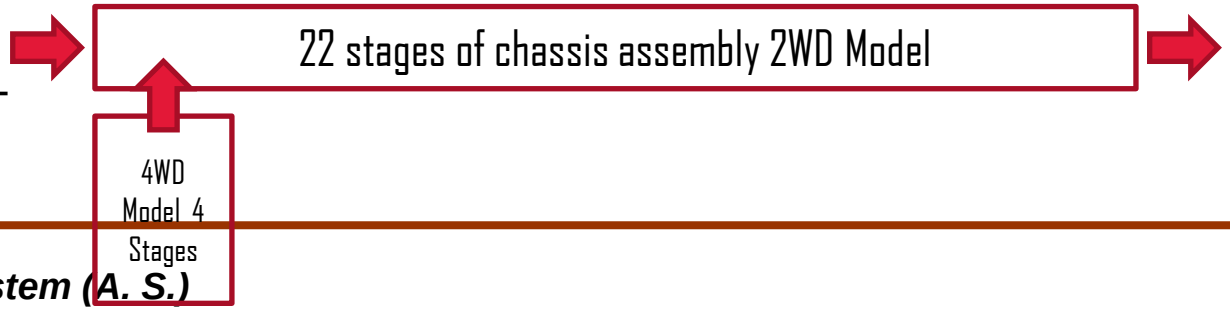
A Explain history of the problem.:

At Scorpio TCF , 4WD model productionised , but due to PFC change apart from existing model , need to complete those activities offline stages -4 nos ,resulted in increase of chassis line lead time

B. Current understanding of the problem(Current system and its capability)

At Scorpio TCF , 4WD models needs additional assy of 4WD unit to mount on chassis .due to which lead time increases , the lead time 66.3 0 min , which need to be reduced .

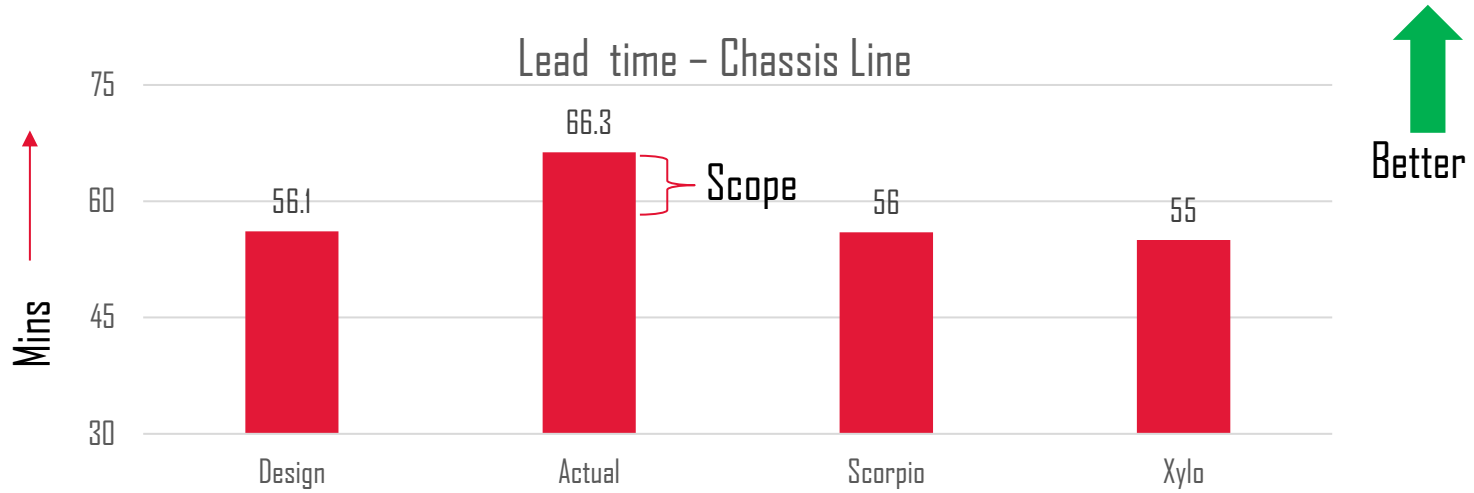
C. Current Flow Chart :-



5. Analysis

List of Possible causes after brainstorming among team members

Benchmark current system with other companies /systems



Inference (Key observations from benchmarking study):

At Scorpio TCF ,4WD model lead time of chassis line 66.3 min , after benchmarking study which clearly shows scope of improvement by 10 mins.

5. Analysis

List of Possible causes after brainstorming among team members

Inference (Key observations from the studies so far)

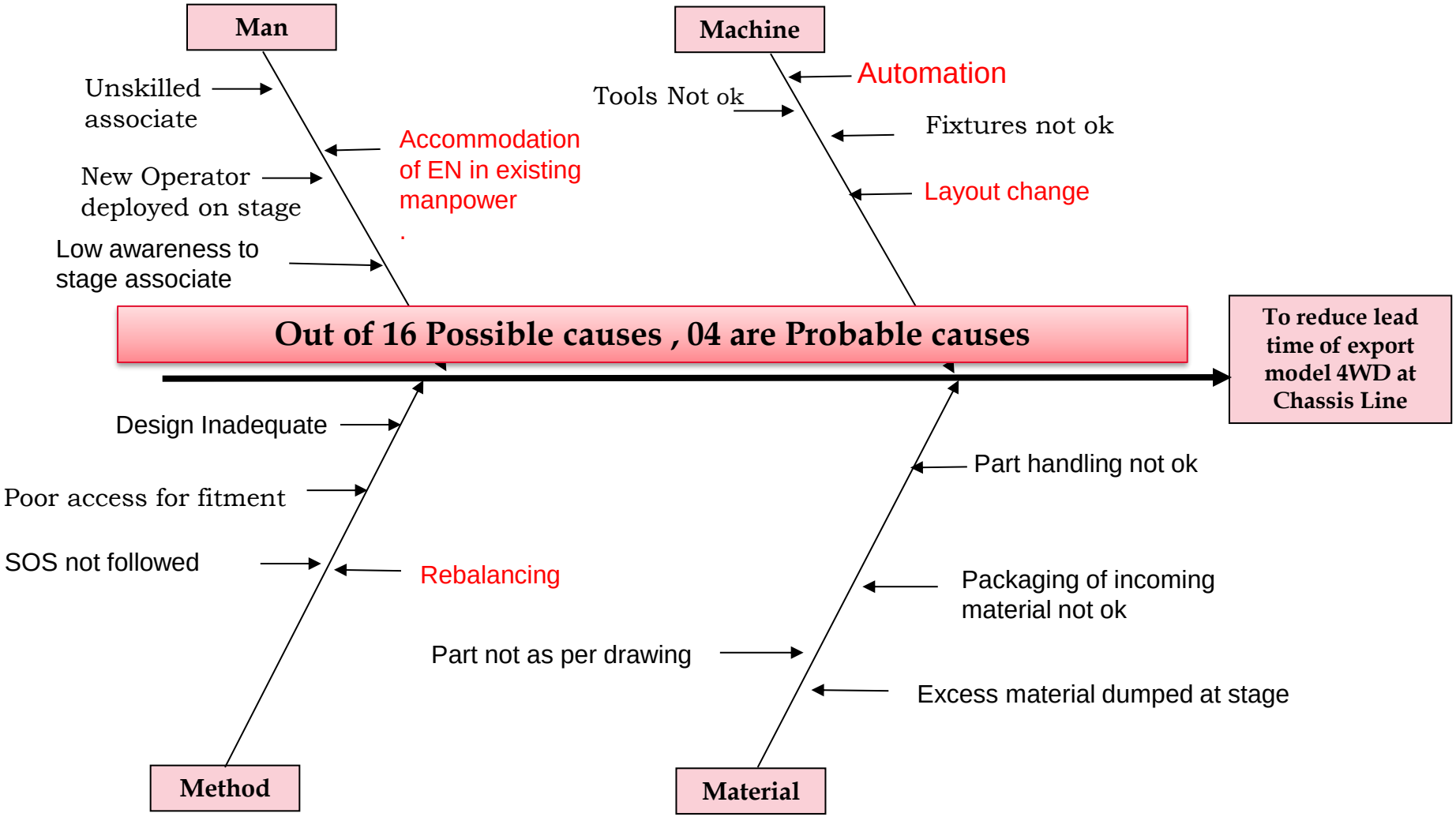
At Scorpio TCF ,4WD model lead time of chassis line 66.3 min needs to be focused and reduced
By definition lead time has 2 factors,

1. Number Of Stages
2. TACT Time

In Our case content of TACT is already freeze as per PFC , but stage can be reduced.

Step 5. Analysis

5.1 Identify possible causes: Fish Bone diagram



5. Analysis

5.1 List of Possible causes

Probable Cause	Taj	Ajit	Santosh	Shreeyash	Darshak	Mandar	Total
Unskilled operator	0	0	1	1	0	1	3
New Operator deployed on stage	1	1	1	1	1	3	8
Accommmodation of EN in existing manpower	9	1	9	9	1	9	38
Low awareness to Stage associate	0	0	3	0	3	3	9
Tools not ok	0	0	0	1	0	1	2
SOS not followed	3	1	1	1	0	3	9
Fixtures not used	3	1	1	1	0	3	9
Layout Change	9	1	9	1	3	9	32
Design Inadequate	0	1	1	3	0	3	8
Poor Access for Fitment	1	1	0	1	0	1	4
Automation	0	9	9	9	3	9	39
Parts not as per drg.	0	0	0	0	0	3	3
Rebalancing	1	3	9	3	9	9	35
Parts Handling	3	3	1	1	0	3	14
Parts Packaging for incoming material	0	1	1	3	0	3	8
Excess material dumped at Stage	0	1	1	3	0	3	8
Fuel Tank Resting Improper due to Resting Foam Size Incorrect	0	0	1	3	0	3	7
CD of the strap on the tank not ok	0	0	0	0	1	3	4

Selection Criteria: 0-No impact, 1- less impact , 3-High impact , 9- Significant impact.
Threshold Value for selecting probable cause is 32.

5. Analysis

5.1 List of Probable causes: Brainstorming

1. **Accommodation of EN in existing manpower**

2. **Layout Change**

3. **Automation**

4. **Rebalancing**

Step 5. Analysis

List of Probable causes identification in Brain storming session

What will be the key points of attack in the new service /system to best achieve our goal? (ie what are the major controllable variables which will influence your objective)

To take work offline to online work – Need to split the work and take online with creation of stage .

What are the main hurdles/ constraints in the project

Model mix productionisation are the main hurdles in the project

Step 5. Analysis

List of Probable causes identification in Brain storming session

Various alternate options to achieve the objective are

a. Accommodation of EN in existing manpower

b. Rebalancing

c. Layout change

d. Automation

Step 5. Analysis

List of Probable causes identification in Brain storming session

Sr. No.	Option identified	Is it Effective?	Is it Feasible?	Does it have risk /consequential Effect?	Is it Economical?	Decision
1	Accommodation of EN in existing manpower	Yes	NO	Low	Yes	Dropped
2	Rebalancing	Yes	Yes	Low	No	Selected Make risk mitigation plan.
3	Layout change	Yes	NO	Medium	Yes	Dropped
4	Automation	Yes	NO	high	No	Dropped

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

a. Accommodation of EN in existing manpower
b. Rebalancing
c. Layout change
d. Automation

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

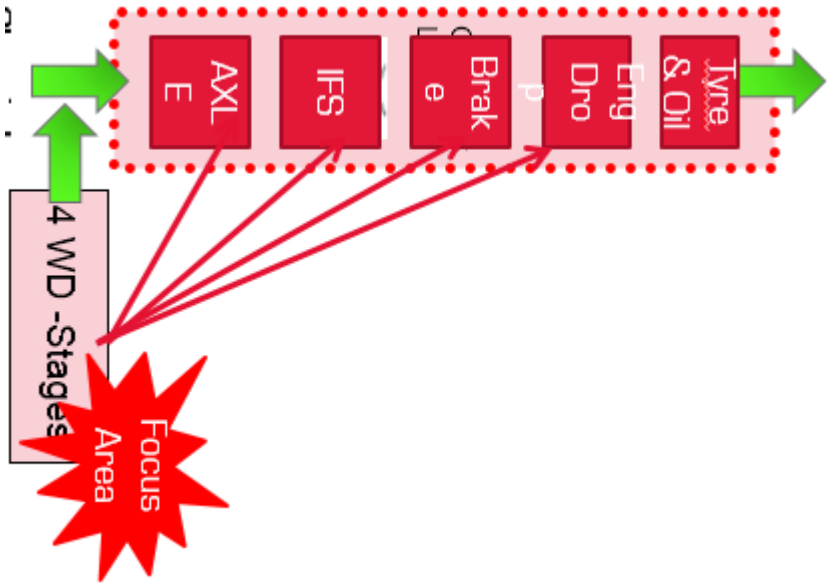
No	Probable Cause	Testing method	Testing and Observations	Conclusion
1	Accommodation of EN in existing manpower	MOST and WC	On observation it was observe that we can not accommodate this in existing manpower	**Hypothesis Invalid**

Existing wc of 30 mins can not get accommodate fully in existing manpower

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
1	Rebalancing	MOST and WC	On observation it was observe that we can rebalance the work	**Hypothesis valid**



Looking at the existing stage utilization , we can rebalance the existing work

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
1	Layout change	Visual inspection and measurement	On observation it was observe that we can not accommodate this in existing layout	**Hypothesis Invalid**

22 stages of chassis assembly 2WD Model

4WD Model 4 Stages

To accommodate 4WD Layout in existing line is very difficult due to increase in number of stages

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
1	Automation	-	On observation it was observe that automation is not economical and suitable in existing method	**Hypothesis Invalid**

Detailed analysis done to identify scope for automation

List of operations can be automised

1. Front Axle drop
2. Bolt tightening

But due to operation constraint we can not atomize both the operations

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Conclusion
1	Accommodation of EN in existing manpower	MOST & WC	Hypothesis invalid
2	Rebalancing	MOST & WC	Hypothesis valid
3	Layout change	Visual inspection and measurement	Hypothesis Invalid
4	Automation	-	Hypothesis invalid

6. Implement the corrective measures

Possible Remedies Identified

5.1 . Actions Decided

Action Plan	Target Dt	Resp.	Status
Concept finalization with all stake holders	05.07.2019	IED	Closed
Area finalization for performing activity at new area.	30.07.2019	Mfg	Closed
Associate Training	10.08.2019	E&T	Closed
Space creation for logistic	15.09.2019	SCM	Closed
Design change as per new requirement	31.10.2019	R&D	Closed
Implementation of improvements with frugal mind set.	30.11.2019	IED	Closed

6. Implement the corrective measures

SN	Line	Stage	Status	Scope For Merging	Scope For Online	Remarks
4	Chassis Line	4WD	Offline	Yes	Yes	Possible only On Chassis Line

At 4WD Sub assy Stage - operations involved

- Chassis Unloading
- Steering Gear Fitment
- Front Axle Drop
- Front Propeller shaft Fitment
- Torsion Bar Fitment
- Pental Hook Fitment

6. Implement the corrective measures

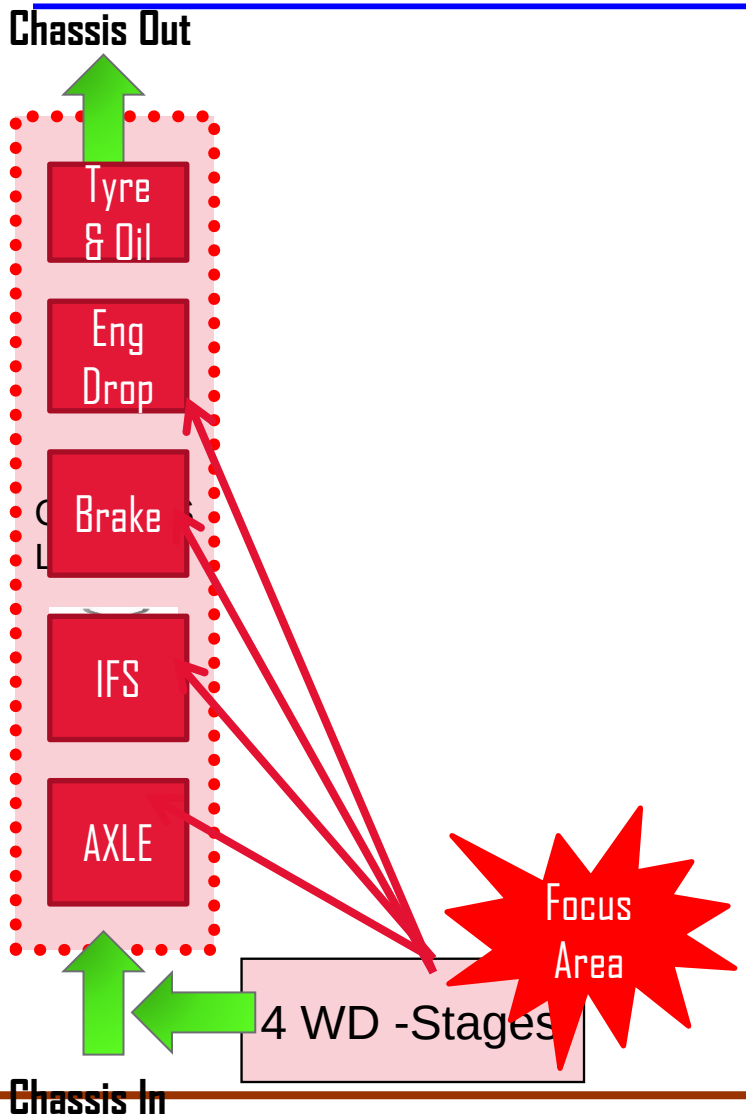
Can we take all operations online ??

Yes ....We can merge on chassis line

SN	Part	From	To	Feasible
1	Chassis Unloading	4WD	Axle Group	Yes – Activity Eliminated
2	Steering Gear Fitment -	4WD	Axle Group	Yes
3	Front Axle Drop -	4WD	IFS – Station 10	Yes
4	Front Propeller Shaft Fitment	4WD	IFS – Station 20	Yes
5	Torsion Bar Fitment	4WD	IFS – Station 30	Yes
6	Pental Hook Fitment	4WD	Brake Group	Yes

6. Implement the corrective measures

Process flow - Multimodel TCF



6. Implement the corrective measures

Title of Kaizen-To implementation of 4WD Axle assy offline to online

Before

Operation Offline Sub assy

Description:

1. 4 associates/ shift required
2. Required 4 stages
3. 75 sq Meter space
4. Push System

After

Operation Online

Description:

1. 3 associates/ shift required
2. Required 1 stages
3. 00 sq Meter space
4. Pull System

6. Implement the corrective measures

Title of Kaizen-To implementation of 4WD Axle assy offline to online

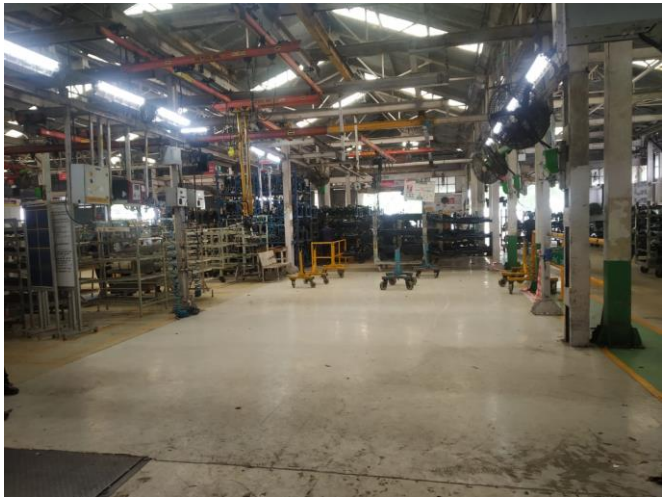
Before

Operation Offline Sub assy

Description:

1. 4 associates/ shift required
2. Required 4 stages
3. 75 sq Meter space
4. Push System

After



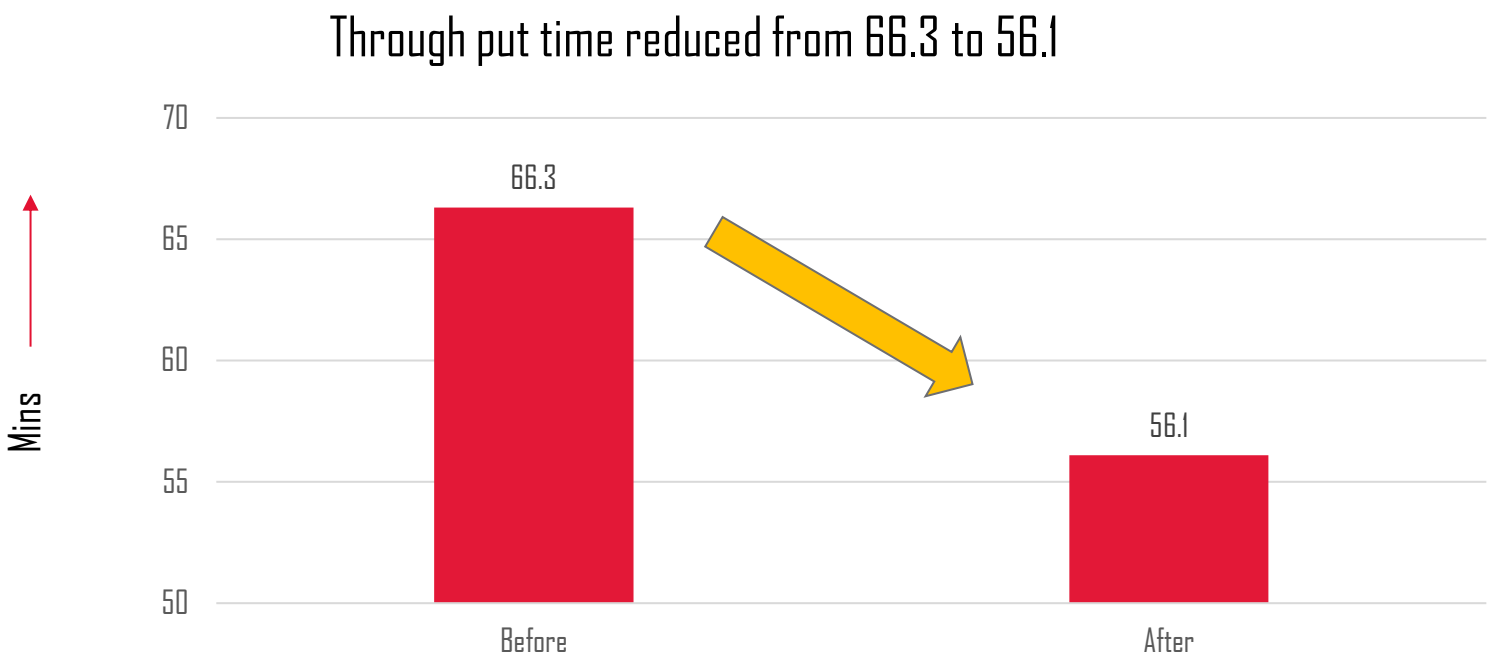
Operation Online

Description:

1. 3 associates/ shift required
2. Required 1 stages
3. 00 sq Meter space
4. Pull System

7. Check & confirm the results & effects

7.1 Verify Results:



Significant Improvement is observed.

Check & confirm the results & effects (Cont.....)

7.2 Tangible & Intangible benefits:

Tangible Benefits -

- Manpower Cost saving / Annum– Rs 1.8 Lakh

In tangible Benefits :

- Part Discrepancy eliminated
- Remote Location Monitoring eliminated
- Moral of the associate improved

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

SN	Action Taken	Actual date	Responsibility	Status
1	PCRN	14.7.19	Datta Jondhale	Closed.
2	SOS Updated	20.12.19	Suryabhan SOR	Closed.
3	Element updated	20.12.19	Rajesh Dhanokar	Closed
4	Process Audit check point added.	25.12.19	Suryabhan SOR	Closed.

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

SN	Document	Actual date	Responsibility	Status
1	PCRN NO -13979	14.7.19	Datta Jodhale	Closed.

25	PCRN/A003/STCF/F20/13979	4WD drive offline to online	P	7/14/2019	8/27/2019	Chassis

PCRN/PPAP Submission - Mahindra & Mahindra Corporate Portal - Google Chrome

https://ep.mahindra.com/irj/servlet/prt/portal/prteventname/Navigate/prtroot/pcd!3aportal_content!2fcom.mahindra.ha_fld_harmony!2fcom.mahindra.ha_fld_hcm!2fc

DISPLAY PCRN

PCR...:* PCRN/A003/STCF/20/13979 | 4WD drive offline to online | Planned closing date:* 31.08.2019 | Actual closing date:

A.Form

PU / Area:* STCF | Cell:* CHASSIS | Date: 14.07.2019

Models Affected: All 4WD

Opn No./Mc No.: | Opn Description: 4WD drive offline to online

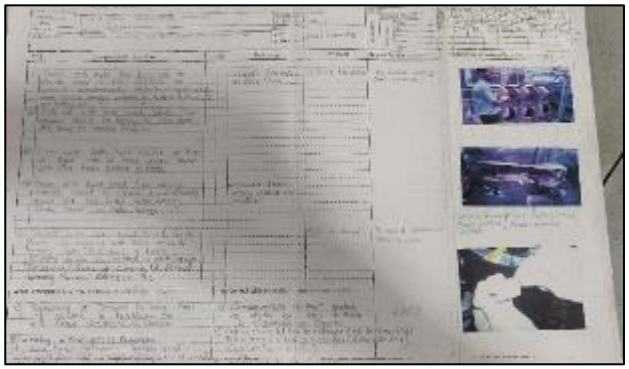
Process Change Trigger

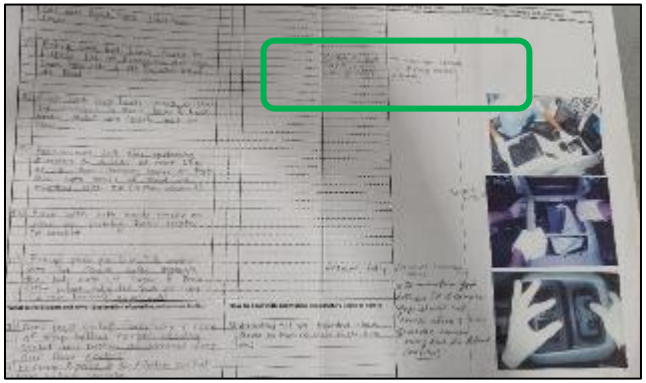
- ☐ Process Parameters
- ☐ Layout Change
- ☐ Production from New M/c
- ☐ Poka Yoka Introduction
- ☐ Inspection & Test Method
- ☐ Reconditioned M/c/ Tooling
- ☐ Method
- ☐ Use of alternate material
- ☐ 25% prodn. Ramp UP than rate quoted in present PPAP
- ☐ Transferred M/c
- ☑ Any Other (Kaizens/ Improvement Projects/ Ingenious Ideas etc.)
- ☐ Tools
- ☐ Product produced after the tooling has been inactive for more than 12 month
- ☐ Change of Vendor for Part, Material, Paint etc
- ☐ Production from tooling and equipment transferred to a different plant location or from additional plant location Production from New or Modified

8. Standardization

8.2 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

SN	Document	Actual date	Responsibility	Status
2	SOS Updated	20.12.19	SOR	Closed.





- Associate respect the main step assembly.
- SOS modified for adequate tightening sequence mentioned

8. Standardization

8.3 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

SN	Document	Actual date	Responsibility	Status
1	Element updation	20.12.19	Rajesh Dhanokar	Closed.

MOST Element Report

Operation Group : X.2 SC_CHS_SUM
Operation No : 6030.2
CASTER BOLT INSERTION & NUT PREFITMENT_SCDC_4WD

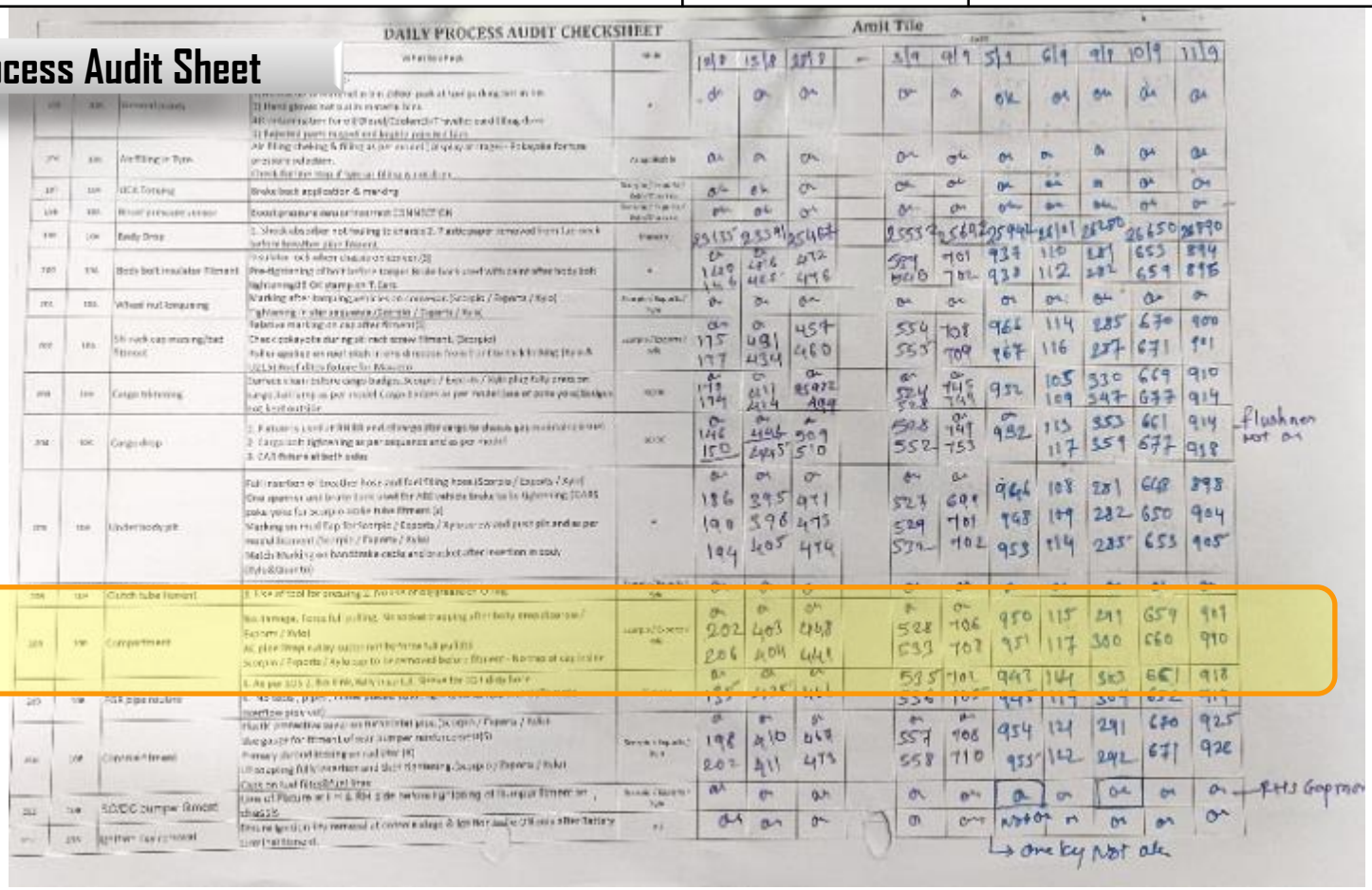
SR NO.	ELEMENTS DESCRIPTION	FRQ.	DIV	MEN	OFFLINE
0.1	**MOVE 3-4 + GRASP SP TOOL & HAMMER + BACK + PLACE ON SP.TOOL BOLT HEAD**	**1.00**	**1.00**	**2.00**	**No**
1	MOVE 3-4 STEPS, GRASP OBJECT , MOVE 3-4 STEPS, PLACE WITH ADJUSTMENT	1.00	1.00	1.00	NO
1	**MOVE 1-2 STEPS AND GRASP BOLT AND CHILSEL**	**2.00**	**1.00**	**2.00**	**No**
1	MOVE 1-2 STEPS, GRASP PART	1.00	1.00	1.00	NO
2	**MOVE 1-2 STEPS AND PLACE BOLT TO INSERT WITH ADJUSTMENT**	**1.00**	**1.00**	**2.00**	**No**
1	MOVE 1-2 STEPS, PART PLACE WITH ADJUSTMENTS	1.00	1.00	1.00	NO

8. Standardization

8.4 Additional Audit & check

6	BWT leaders supervisory check sheet updated.	25.12.19	sor	Closed.

Process Audit Sheet



9. Summarize & Plan future actions

9.1 Conclusion:

Achieved lead time reduction for export model thru offline to online 4WD assy

9.2 Horizontal Deployment:

NA.

9.3 Lessons Learnt:

Things Gone Wrong :

- Accommodation of single tackle use not possible for both the model

Things Gone Right :

- Use of associate feedback helped in analysis of root cause
- Process walk though

Thank you

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